GENAI BASICS • LESSON 06 • MODULE 2: MULTIMODALITY

# Speech and Audio

## Voice-over, sound and music for a video, and how to check the result

Text-to-speech, speech-to-text, voice agents, music. ElevenLabs, OpenAI, Google, Suno, open-weight models. Data as of October 2026.



EXAMPLE

# The L05 courier video with voice-over, sound and music




## 3 lines

9.3 s of speech from an **open-weight** text-to-speech model (downloadable, runs locally) on an ordinary laptop

## 3 tracks

voice + the sound Veo generated with the video + music

## −14.4 LUFS

loudness like Spotify and YouTube (their norm is -14). Details in step 5

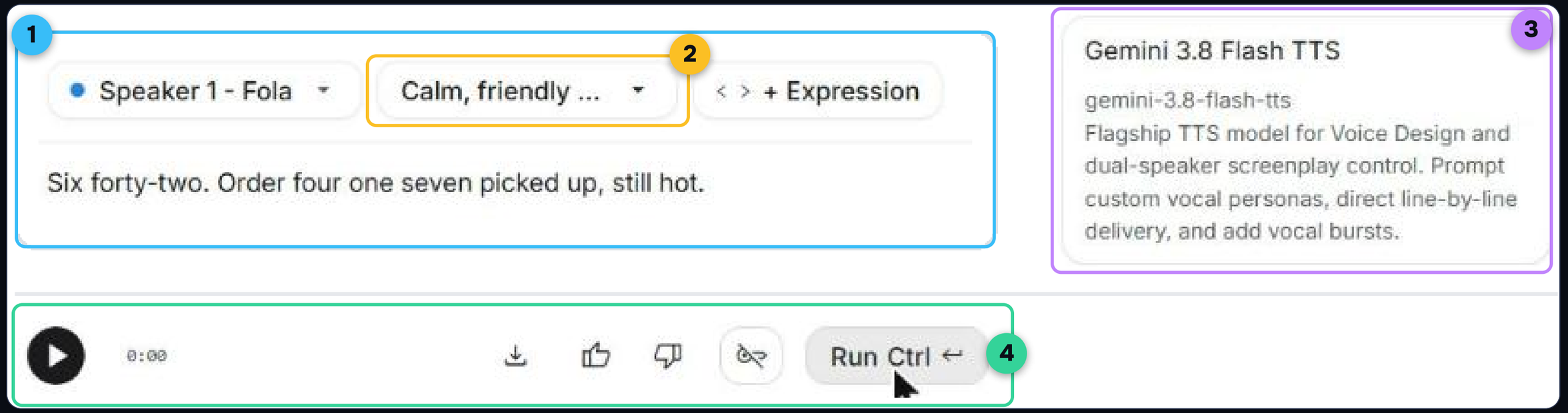
LESSON PIPELINE 1 Script 2 Spoken text 3 Voice 4 Effects and music 5 Mix 6 Check

Can’t hear the video? Tell us in the chat.

WHERE TO GENERATE

# Start in a web studio, automate with the API

WEB STUDIO · GOOGLE AI STUDIO → GENERATE SPEECH · FREE




1. **Text** of the line
2. **Style** separately: “calm dispatcher”
3. **Model** Gemini 3.8 Flash TTS
4. **Result**: 5 s, download WAV

API · THE GET CODE BUTTON → READY PYTHON, THE KEY LINE:

```
client.models.generate_content(model="gemini-3.8-flash-tts", contents=...)
```

Two more paths: a chat app (to try things) and open-weight models on your machine (Kokoro, Piper: how our example was made).

TASK MAP

# Four audio AI tasks: what goes in and what comes out

**text** → **voice**

## Text-to-speech (TTS)

video voice-over, notifications, audiobooks. **Today: the courier narration.**

In our example: **Kokoro (EN) and Piper (RU)**

**voice** → **text**

## Speech-to-text (STT)

subtitles, meeting notes, call search. Also called automatic speech recognition (ASR). **Today: checking the voice-over.**

In our example: **Whisper small, the voice-over check**

**voice** → **voice**

## Speech-to-speech

voice agents, live translation (real time), dubbing

In our example: **— (agents are built in module 5 (L11–L13))**

**text** → **music, effects**

## Music and sound effects (SFX)

beds, jingles, effects. Plus the sound a **video model** makes by itself (L05).

In our example: **MusicGen, the music bed**

HOW IT WORKS

# The model guesses how text sounds, so it trips on numbers and names

**Text**
“Six forty-two.”

→

**Model**
guesses chunks of sound

→

**Sound chunks**
audio tokens, 12–50 steps per second

→

**Decoder**
rebuilds the waveform

→

**Sound**
24,000–48,000 numbers per second

LINE 1 OF OUR SCRIPT, REAL WAVEFORM · 3.9 S



**Compression:** a second of audio is tens of thousands of numbers. The model works with a short summary of a few dozen chunks; the decoder expands it back into a waveform.

**What is not inside:** a rulebook for reading. The model learned how text usually sounds, so it most often errs on **numbers, names and abbreviations**.

BRIDGE FROM L05

# Veo makes up its own lines: exact words need a separate voice

**Shot 1, cafe** prompt has no dialogue



▶ audio in the HTML version

HEARD IN VEO'S AUDIO

"Here's your order. Thank you. Have a good day."

**The L05 failure clip** prompt has no dialogue either



▶ audio in the HTML version

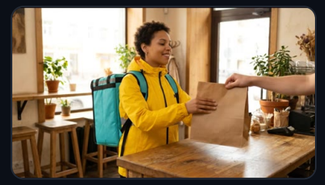
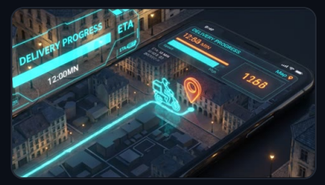
HEARD IN VEO'S AUDIO

"Continue straight for 200 meters, then your destination is on the left."

**Provides:** street noise, rain, footsteps, in sync with the picture. But you pay for it: Veo 3.1 on Vertex AI is $0.40/s with audio, $0.20/s without.

**Does not provide:** exact words, one voice across shots, separate tracks. The model **makes up lines** by itself.

# Voice-over script: text, timing and a word budget for each shot

| SHOT | TIME | VOICE-OVER | WORDS / LENGTH | SOUND EFFECTS |
|---|---|---|---|---|
|  | 0–8 s | “6:42. Order 417 is picked up, still hot.” | 11 / 3.9 s | cafe: room noise, cups |
|  | 7.5–15.5 s | “Rain on the avenue. The route skips the bridge.” | 9 / 2.7 s | rain, tyres on puddles |
|   | 15–23 s | “12 minutes to the door. Anna is on her way.” | 10 / 2.7 s | soft UI sound |
| Music: one bed for all shots, lo-fi, no vocals | | | | |

## ≈ 3.2 words/s

our voice: faster than plan; it fits only because the lines are short. Plan: **2–2.5 words/s**

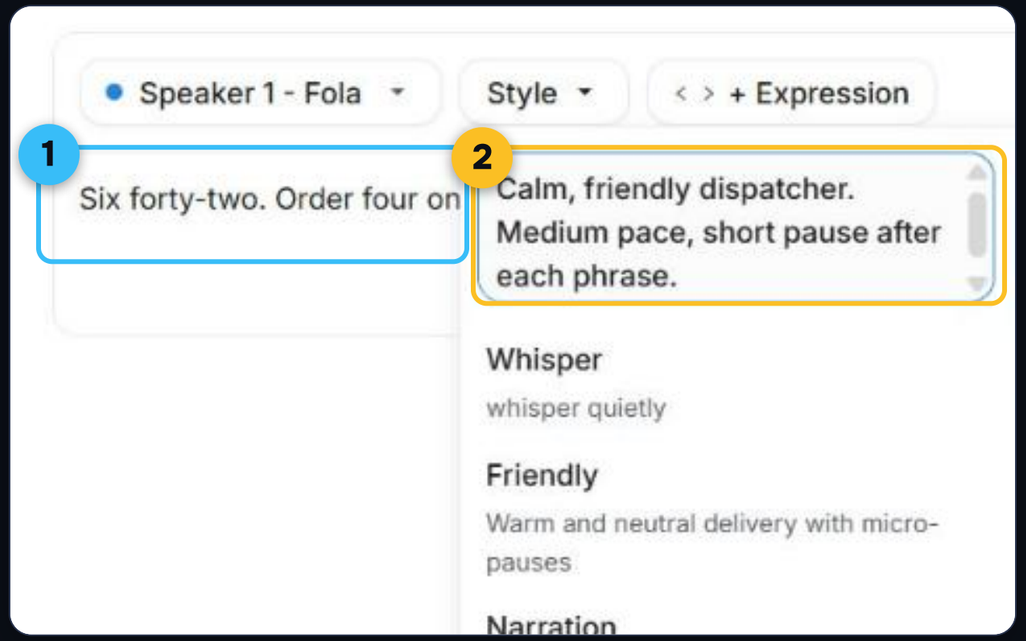
**An 8 s shot ≈ 6 s of speech** (about 1 s of silence at each cut) **× 2–2.5 words/s = 12–15 words.** If it does not fit, cut the text; do not speed up the voice.

# Spoken text: numbers as words, style as a separate instruction

WHAT THE VIEWER SEES → WHAT THE MODEL READS

| SCRIPT | SPOKEN TEXT | WHY |
|---|---|---|
| 6:42 | Six forty-two | a time, a score or a ratio? |
| 417 | four one seven | an order ID: digit by digit, not “four hundred seventeen” |
| 12 min | twelve minutes | abbreviations may be read letter by letter |

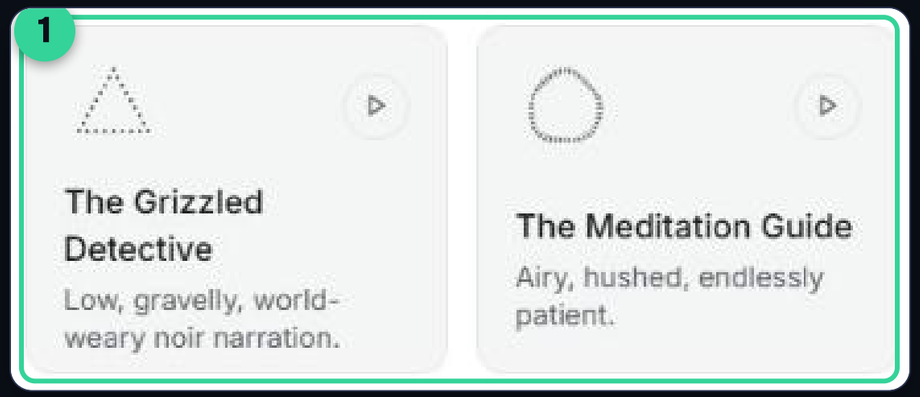
STYLE GOES IN A SEPARATE INSTRUCTION




1. **Text**: only what must be spoken
2. **Style** in its own field (Google AI Studio). ElevenLabs v3 uses tags in the text: `[whispers]`

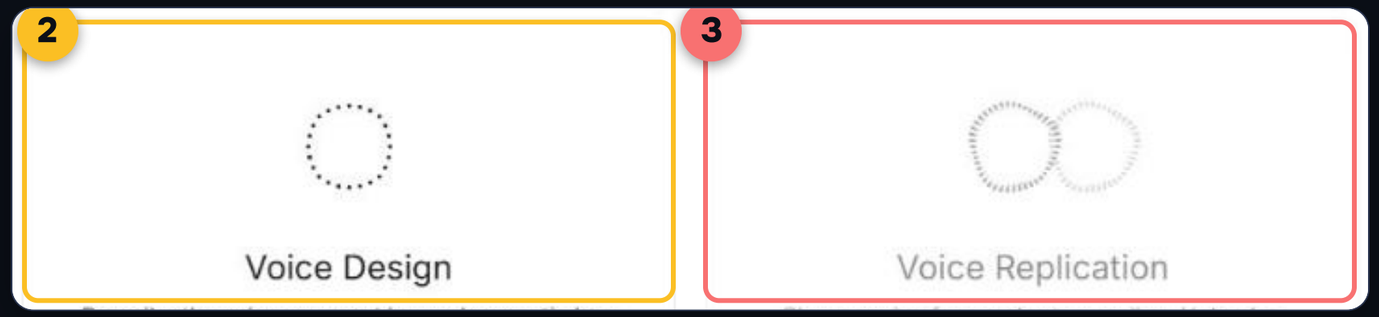
# Three sources of a voice: library, description, clone

GOOGLE AI STUDIO · VOICE LIBRARY




1. **Library voice**: covered by the service licence
2. **Designed voice**: “young, slightly husky”; check it doesn’t sound like a celebrity
3. **Clone** from a recording: the service asks you to prove the voice is yours

CREATE NEW VOICE · TWO NEW PATHS




Voice Design: describe age, timbre, accent and pace in words. Replication: upload a recording and verify the voice is yours.

> **Before you commit:** run the hardest words of the script (names, numbers) through the voice. Reject a voice that mispronounces them.

# Sound effects and music: the licence matters most

**Our music bed** MusicGen-small · 23 s



▶ audio in the HTML version

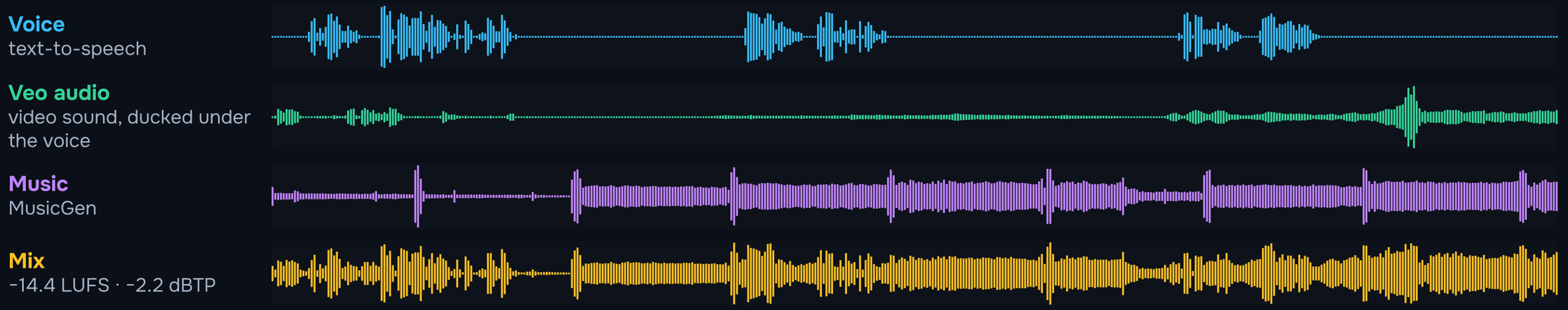
PROMPT

```
calm lo-fi hip hop beat, soft electric piano, light rain
mood, 85 bpm, no vocals
```

> MusicGen weights: **Creative Commons Attribution-NonCommercial (CC BY-NC 4.0)** licence. Fine for a lesson, not for a client.

| SERVICE | LENGTH | COMMERCIAL USE |
|---|---|---|
| **Suno v6**<br>songs | song | paid plans only |
| **Lyria 3.5**<br>Gemini app, Gemini API | ~2 min | per Google's terms; SynthID mark; API $0.08 per song |
| **ElevenLabs Music**<br>music | 3 s – 5 min | "nearly all", per plan |
| **ElevenLabs Sound Effects**<br>sound effects | ≤ 30 s | per plan |

# Mixing: voice on top, background ducked, loudness normalised




- **Ducking:** the background drops automatically while the voice speaks.
- **Loudness −14 LUFS** (a loudness unit that follows the ear): the Spotify and YouTube norm; TV uses −23.
- **Peak in the final file no higher than −1 dBTP**, otherwise encoding makes it crackle. Set the limiter to −2 for headroom, see right.

**No code, Audacity (free):** import the three tracks → background: *Effect* › *Auto Duck* → *Loudness Normalization* −14 LUFS → *Limiter* −2 dB (headroom: AAC in the video adds ~0.5 dB) → WAV into CapCut or Clipchamp.

# Speech-to-text found errors we missed by ear

SCRIPT

6:42. Order 417 is picked up, still hot. Rain on the avenue. The route skips the bridge. 12 minutes to the door. Anna is on her way.

WHAT WHISPER HEARD (SMALL MODEL, LAPTOP)

~~Here's your order.~~ 642. Order 417 is picked up, still hot. Rain on the Avenue, the route skips the bridge. 12 minutes to the door, Anna is on her way.

- **“Here’s your order.”**: the mix: Veo’s invented line leaks through
- **“642” instead of “6:42”**: speech-to-text formatting; the voice said it correctly

**18.5%**

word error rate (WER) = (substitutions + deletions + insertions) / words in the script

**3 / 4** automatically → **4 / 4** after listening

critical terms correct (time, order number, minutes, name)

**No code:** ElevenLabs Speech to Text, Google AI Studio (prompt “Transcribe verbatim. Do not correct mistakes.”), Buzz (Whisper desktop app).

LOCAL MODELS

# Open-weight models: $0 per minute, paid for in hardware and time

Hugging Face
model card and weight files
→
Your disk
~/.cache/huggingface
→
Python package
qwen-tts, kokoro-onnx
→
Processor (CPU) or graphics card (GPU)
computes the audio
→
WAV file
nothing leaves the machine

| MODEL | FILES | RUNS ON | IN THIS LESSON |
|---|---|---|---|
| **Kokoro-82M · Piper**<br>text-to-speech | 0.36 · 0.06 GB | laptop processor | our EN · RU voice |
| **Whisper small**<br>speech-to-text | 0.5 GB | laptop processor | our check |
| **Qwen3-TTS 0.6B**<br>text-to-speech + cloning | 2.5 GB | processor: ≈8–9× slower than real time | slide 16 |
| **Qwen3-TTS 1.7B**<br>text-to-speech + cloning | 4.5 GB | graphics card with 8 GB or more | better quality |

## $0 per minute

no per-character or per-minute bill, but you pay with **hardware, electricity and setup time**

> Why: **data stays on the machine**, no quotas, works offline. Cost: quality below the best services; updates and licences are your job.

MARKET • OCTOBER 2026

# Audio models by task: commercial and open-weight

**Text-to-speech (TTS)**
text → voice

- ElevenLabs v4 / v3
- Gemini 3.8 Flash TTS
- OpenAI gpt-4o-mini-tts (until 6 Jan 2027)
- **Kokoro-82M** Apache-2.0
- **Qwen3-TTS** Apache-2.0

**Speech-to-text (STT, ASR)**
voice → text

- OpenAI gpt-transcribe
- ElevenLabs Scribe v2
- AssemblyAI Universal-3.5 Pro
- **Whisper large-v3-turbo** MIT
- **Qwen3-ASR** Apache-2.0

**Speech-to-speech**
voice agents, translation

- OpenAI gpt-realtime-2.1 / gpt-live-1
- Gemini Live API
- ElevenLabs Agents

**Music and sound effects**
text → music, effects

- Suno v6
- Lyria 3.5
- ElevenLabs Music / Sound Effects
- **MusicGen** CC BY-NC

**Speech-to-text accuracy (English):** the best systems ≈ 4–5% word error rate, Whisper large-v3 ≈ 7–8%. Hugging Face Open ASR Leaderboard; the figures move, recheck them.

Green border: open-weight model and its licence. **MIT, Apache-2.0:** commercial use allowed, keep the licence notice. **CC BY-NC:** non-commercial only. Voices and datasets can have their own licences.

BREAK

# 5 minutes for coffee and resting your ears

05:00

Next: an agent clones a voice, cost, acceptance, risks and homework.

HANDS-ON WITH AN AGENT

# A coding agent installs Qwen3-TTS locally and clones a voice

PROMPT FOR A CODING AGENT: CODEX IN CHATGPT OR ANTIGRAVITY (FREE, WITH LIMITS), CLAUDE CODE (PAID)

```
Install the open-weight text-to-speech model Qwen3-TTS
(Qwen/Qwen3-TTS-12Hz-0.6B-Base, Apache-2.0) in C:\ai\qwen-tts-
demo and clone MY voice.
0. My sample: my_voice.m4a from Windows Sound Recorder, 10 s, and
my_voice.txt with its exact text. Convert the sample to 24 kHz
mono WAV with ffmpeg.
1. Check the graphics card and its memory, system memory, free
disk (~5 GB). Tell me what fits: 0.6B or 1.7B, graphics card or
processor.
2. Create a separate Python environment there; install qwen-tts
and soundfile.
3. Download the model from Hugging Face and show where the files
are.
4. Write clone.py and say in my voice, in English: “Six forty-
two. Order four one seven is picked up, still hot.” → line1.wav.
5. Run it, report the time, upload nothing to the internet.
```




**Sample** synthetic Kokoro voice, 5.8 s

▶ audio in the HTML version

**Clone** line 1: 4.6 s of speech in 42 s on a processor

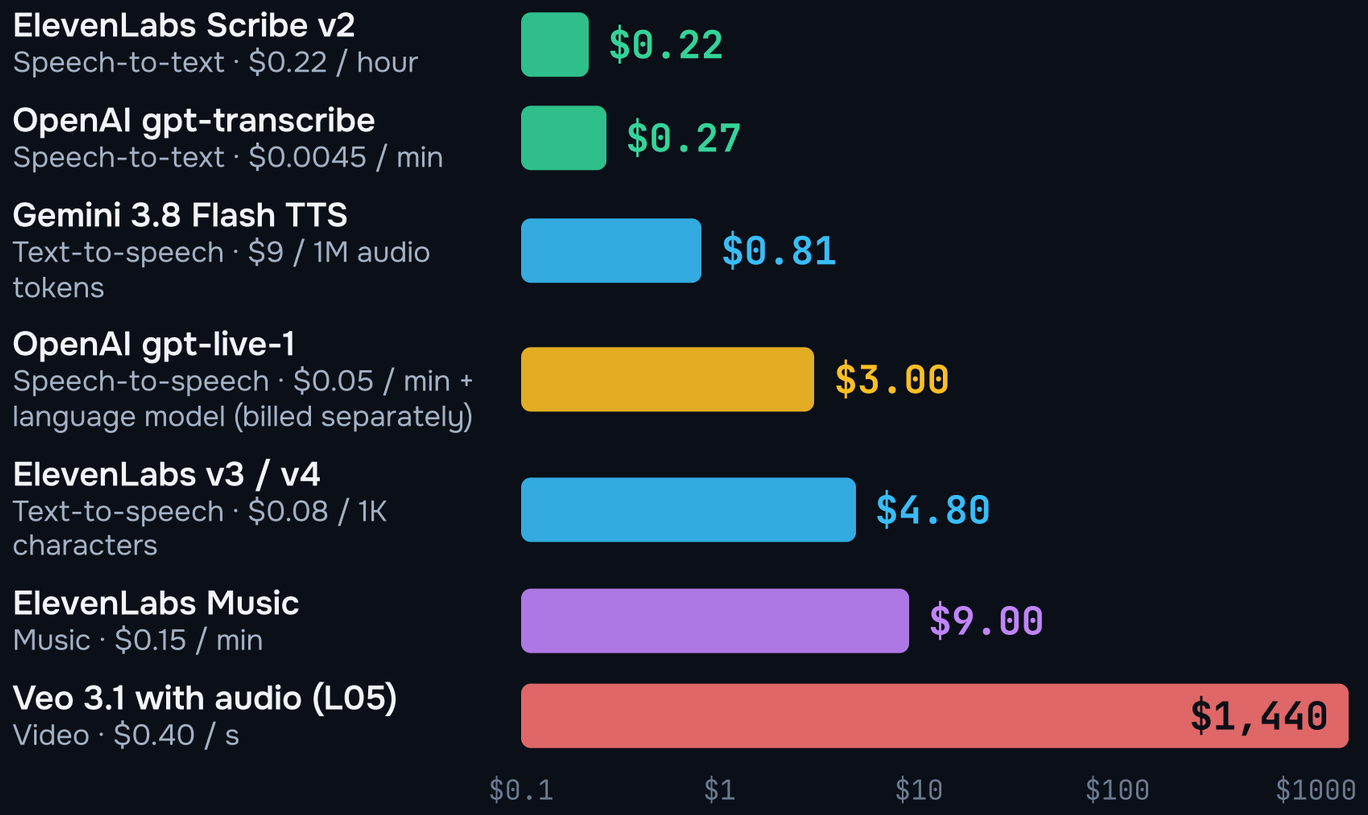
▶ audio in the HTML version

**Your own voice only.** The model checks nothing: consent is your responsibility (slide 19).

COST

# An hour of audio costs cents to dollars; an hour of video, over a thousand

Price of **one hour**, log scale (each tick ×10). 1 minute of speech ≈ 1,000 characters. List prices as of 16 October 2026.




## ≈ 1 cent

voicing our video: 154 characters on ElevenLabs v3; $0 for us on open models

## $4.80 = 12 s

an hour of ElevenLabs voice costs as much as 12 seconds of Veo 3.1 video (L05: $0.40/s)

ACCEPTANCE

# Both our versions failed acceptance: here is the checklist

The same items are the homework rubric. On the right, the real result for our version.

| | | |
|---|---|---|
| 1 | **Words** match the script: speech-to-text + critical terms list | ✗ |
| 2 | **Pronunciation** of names, numbers, terms: checked by ear | ✓ |
| 3 | **Timing:** each line fits its shot and is not cut by a transition | ✓ |
| 4 | **Mix:** voice is clear, background ducked, no stray speech | ✗ |
| 5 | **Loudness** −14 ± 1 LUFS; true peak ≤ −1 dB (dBTP) | ✓ |
| 6 | **Rights:** voice and music licence, consent, “AI-generated” label | ! |

## −14.4 LUFS

peak −2.2 dBTP. No code: Youlean Loudness Meter or the DaVinci Resolve loudness meter; read Integrated and True Peak

> **Rejected:** Veo’s line “Here’s your order” leaks into the start. Fix: mute the shot 1 audio for 0–3 s, then repeat item 1.

Homework evidence: 1 → transcript.txt and critical-terms table · 2, 4 → by-ear notes with timecodes · 3 → script.md · 5 → meter screenshot · 6 → licence table and “AI-generated” label

RISKS

# Risks and voice consent: cloning, labelling, law

- **Library or designed voice**
  allowed under the service licence, with an “AI-generated” label
- **Your own voice, a colleague’s clone**
  written consent: purpose, where, how long, how to withdraw
- **A client’s, manager’s or public figure’s voice**
  not without written consent and legal review; voices are used for fraud

HOW SERVICES CHECK CONSENT FOR A CLONE

- **OpenAI, Google:** the voice owner records a consent phrase. **Azure:** access on application.
- **ElevenLabs Professional:** the owner reads a verification text aloud; Instant: a checkbox only.
- **Open-weight models** (Qwen3-TTS, Chatterbox): clone from 3–10 s, **no checks at all**.

**EU AI Act, Art. 50:** from 2 August 2026, synthetic audio and deepfakes must be labelled; fines up to €15M or 3% of turnover. **OpenAI** requires telling listeners the voice is AI-generated. **Google** embeds an invisible SynthID watermark in audio from its own models.

PRACTICE

# Homework 6: voice your L05 video



## Your L05 video

Track A (product) or B (courier, Glovo case). If you have no video, use the lesson's courier video. No sound in your video: build the background from effects or music.

```
genai-homeworks/L06/
├── script.md       # text, spoken text, timing
├── audio/          # lines, bed, mix
├── final_voiced.mp4
├── transcript.txt # speech-to-text output
├── audio_spec.md
└── audio_qa.md
```

1. **Script**: 3 lines, word budget, critical terms → `script.md`
2. **Spoken text**: numbers and names as words, a style instruction
3. **Voice** in two no-code services (e.g. ElevenLabs and Google AI Studio), keep the better one
4. **Background**: video model audio, effects or music, with its licence
5. **Mix** into `final_voiced.mp4`: -14 ± 1 LUFS, a meter screenshot
6. **Check**: `transcript.txt`, slide 18 checklist → `audio_qa.md`, cost → `audio_spec.md`

Reviewer: **ihar_rubanovich@epam.com**. Your own voice or library voices only; do not clone anyone else's voice.

SUMMARY

# Lesson summary

1. **Script with a word budget**
   2–2.5 words per second, a second of silence at each cut
2. **Spoken text kept separate**
   numbers, times, names as words; style as a separate instruction
3. **A voice with a licence and consent**
   a clone only with the owner’s written consent
4. **Background with a known licence**
   video audio, effects or music; MusicGen and free Suno are non-commercial
5. **Mix to −14 LUFS**
   voice on top, background ducked, true peak ≤ −1 dB (dBTP)
6. **Check with speech-to-text and by ear**
   word error rate + critical terms



**Capstone 1 “Voice Brief → Image”** starts after this lesson: a no-code Colab, 30 of the 100 course points.

**Lesson 07: the Microsoft Copilot ecosystem.** From generating content to AI inside everyday work tools.